PHYS7101: NUMERICAL METHODS AND DATA ANALYSIS · LECTURE 7

From series to shapes

Phase space, linear algebra and decompositions

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Where we left off

From earlier lectures

- Limit cycle: a self-sustained oscillation (L2)
- Linear systems: decay, growth, oscillation read from eigenvalues (L2)
- Normal equations, Jacobian, conditioning κ (L1, L3)
- Autocorrelation; the order carries the information (L6)

Today

A Phase space: the third view of a time series; fixed points, limit cycles, eigenvalues

B Solving linear systems: $Ax = b$, conditioning

C Decomposing data: correlation, SVD, PCA

D Outlook: other decompositions

Lecture 6: two views of a time series, in time and in frequency. Today the third: phase space.

Each decomposition assumes something about the shape of the set of points: look at the points first. Running example: the van der Pol oscillator of Lectures 2 and 6, $\mu = 2$.

PART A

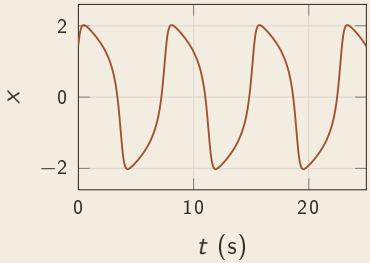
Phase space

A series as a path through the states of the system.

One series, three views

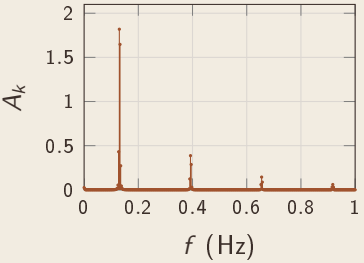
Problem: a plot of $x(t)$ shows one value at a time. How do we see the state of the system?

time domain (L6)



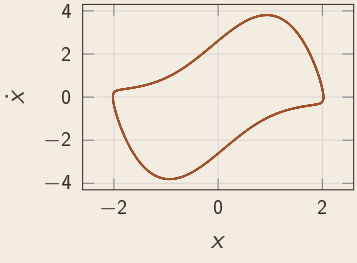
period, shape, slow and fast parts

frequency domain (L6)



$f_0 = 1/P$ and its harmonics

phase space (today)



the state: one closed curve

$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$. Right: each moment plotted as velocity $\dot{x}$ against position x – every period traces the same closed curve.

Data: simulated throughout this lecture – the van der Pol oscillator $\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$, $\mu = 2$, solved with RK4 (L1–L2), sampled every 0.05 s; measurement noise added where stated.

State, phase space and trajectory

Problem: what do we need to know at one moment to predict the future?

- **State:** the numbers that, with the equations, fix the future. For a second-order equation such as van der Pol: position and velocity, $(x, \dot{x})$.
- **Phase space:** the space of all possible states; here the plane $(x, \dot{x})$.
- **Trajectory:** the path of the state in time, starting from the initial condition (Lecture 1: an initial-value problem).

Autonomous and nonautonomous

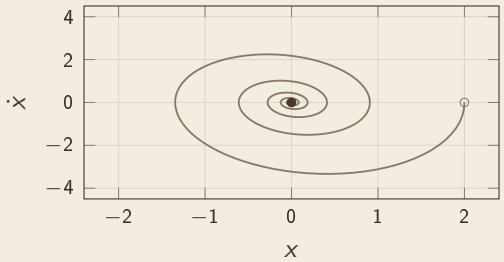
Autonomous system (L2): the state fixes the future, so through each point of phase space passes exactly one trajectory – trajectories never cross.

Nonautonomous system: the same phase space, but the rule changes with the parameters in time – the same state can lead to different futures; fixed points and limit cycles move (slide 7).

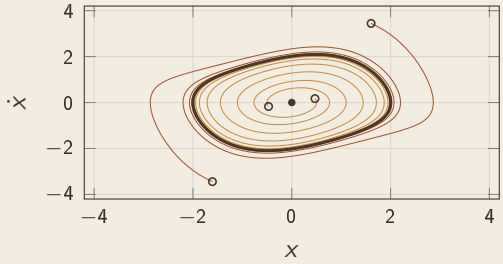
Fixed points and limit cycles

Problem: where do trajectories go in the long run?

Mass–spring–damper (L2): to a fixed point



van der Pol, $\mu = 0.3$: to a limit cycle

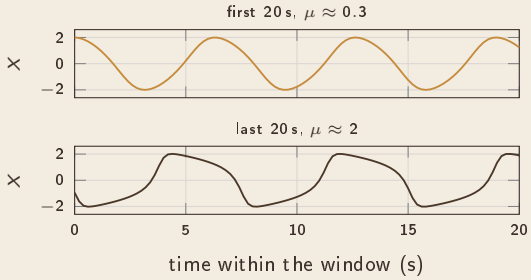


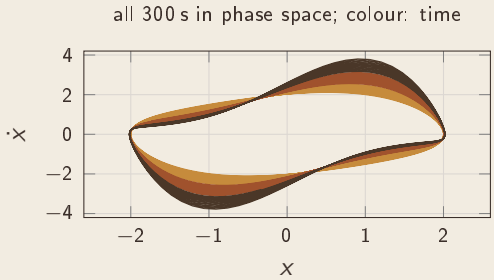
Fixed point: a state that does not change, $\dot{x} = 0$. **Limit cycle**: an isolated closed trajectory. **Attractor**: the set of states that trajectories approach in the long run – left a *point attractor*, right a *cycle attractor*. Open circles: starts. Right: from any start except the fixed point at the origin, the trajectory ends on the same cycle; its size and shape are set by the system. Larger μ: a distorted cycle ($\mu = 2$, slide 4), the same behaviour.

Data: simulated with RK4: mass–spring–damper ($c = 0.5$, $k = 4$, $m = 1$) and van der Pol ($\mu = 0.3$). Lecture 7 – Phase space and decompositions

When the parameters change in time

Problem: what does phase space show for a nonautonomous system (L2)?





μ rises slowly from 0.3 to 2. Same phase space, but the rule changes: the cycle changes shape (largest $\dot{x}$ 2.5 → 3.8, period 6.4 → 7.6 s). Right (ochre: 0–100 s, red: 100–200 s, dark: 200–300 s): no single closed curve – states from different times are mixed.

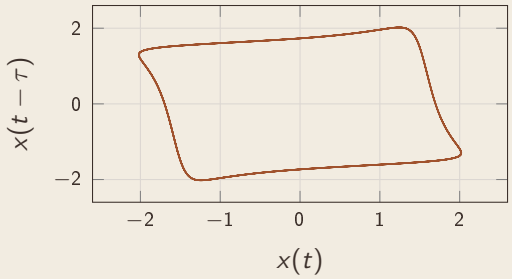
Phase space suits autonomous dynamics; nonautonomous change must be followed in time (L6).

Data: van der Pol with $\mu(t) = 0.3 + 1.7\,t/300\,\mathrm{s}$, RK4, 300 s.

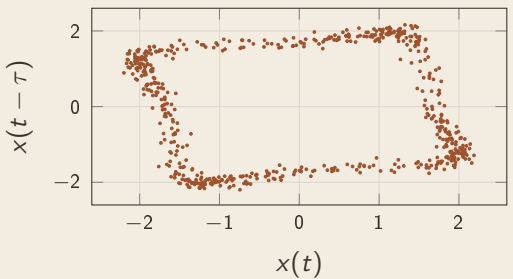
Phase space from one measured series: delay coordinates

Problem: usually only one quantity is measured, e.g. $x(t)$, and $\dot{x}$ from data is noisy (Lecture 4). Can we still draw the phase space?

without noise



with measurement noise, $\sigma = 0.1$

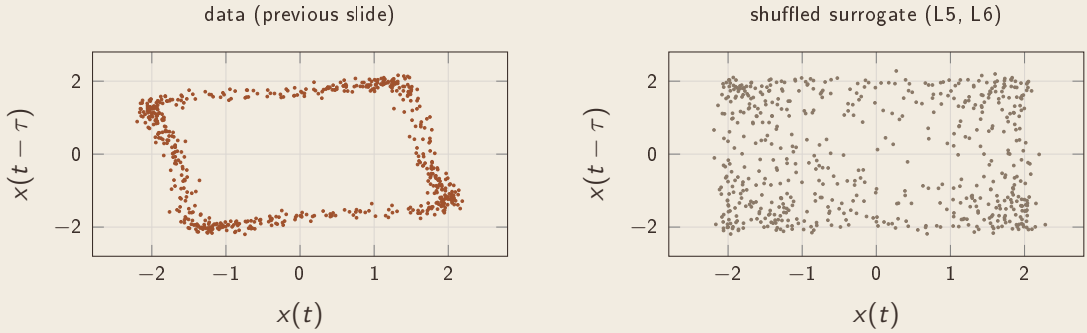


Delay coordinates: plot $x(t)$ against $x(t-\tau)$. Choose τ where the autocorrelation (L6) first crosses zero: here $\tau = 1.95\,\mathrm{s}$, about $P/4$. The closed curve reappears; its shape is distorted, but it is still one closed curve. For more complex systems, use more delays: $(x(t), x(t-\tau), x(t-2\tau), \dots)$.

Data: van der Pol of slide 4, 200 s; right: plus independent Gaussian noise, $\sigma = 0.1$.

The shape is in the order

Problem: is the ring a property of the values, or of their order?



The same values, shuffled: same mean, s and histogram, but no ring. The shape in phase space is in the order of the values, as the rhythm was in Lecture 6. Noise blurs the ring; it does not remove it.

Data: the noisy series of slide 8; right: the same values in random order.

Eigenvalues and eigenvectors

Problem: for a linear system $\dot{\mathbf{x}} = A\mathbf{x}$, how do we know whether it decays, grows or oscillates without solving it?

An **eigenvector** $\mathbf{v}$ of A is a direction that A only stretches: $A\mathbf{v} = \lambda\mathbf{v}$, with the **eigenvalue** λ (real or complex). Along it, $\dot{\mathbf{x}} = \lambda\mathbf{x}$, so

$$\boxed{\mathbf{x}(t) = \sum_i c_i\, e^{\lambda_i t}\, \mathbf{v}_i} \qquad (c_i \text{ from the initial condition}).$$

Eigenvalue $\lambda = \alpha + i\omega$	Motion along $\mathbf{v}$
$\alpha < 0$	decays at rate $\lvert\alpha\rvert$
$\alpha > 0$	grows: the state leaves
$\omega \neq 0$	oscillates at angular frequency ω

A nonlinear system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ near a fixed point: replace A by the Jacobian $J_{ij} = \partial f_i/\partial x_j$ there (L3).

Eigenvalues at the two fixed points

Problem: do the eigenvalues predict the motion we saw on slide 6?

Mass–spring–damper, $c = 0.5$, $k = 4$

$\dot{x} = v$, $\dot{v} = -kx - cv$:

$$A = \begin{pmatrix} 0 & 1 \\ -4 & -0.5 \end{pmatrix}, \quad \lambda = -0.25 \pm 1.98\, i$$

Decays at rate $0.25\,\mathrm{s}^{-1}$, oscillates at $\omega = 1.98$ rad/s: exactly $\omega_d = \sqrt{k/m - (c/2m)^2}$ of L2. Negative real parts: the fixed point is an **attractor** (the inward spiral).

van der Pol at the origin

Jacobian at $(0, 0)$:

$$J = \begin{pmatrix} 0 & 1 \\ -1 & \mu \end{pmatrix}, \quad \lambda = \tfrac{1}{2}\left(\mu \pm \sqrt{\mu^2 - 4}\right)$$

$\mu = 0.3$: $\lambda = 0.15 \pm 0.99\, i$, grows while oscillating (slide 6). $\mu = 2$: $\lambda = 1, 1$, leaves without oscillating. Positive real parts: the origin repels; the state settles on the limit cycle, as for $|x| > 1$ the damping turns positive.

Eigenvalues give the local motion near a fixed point; the limit cycle is a global, nonlinear property.

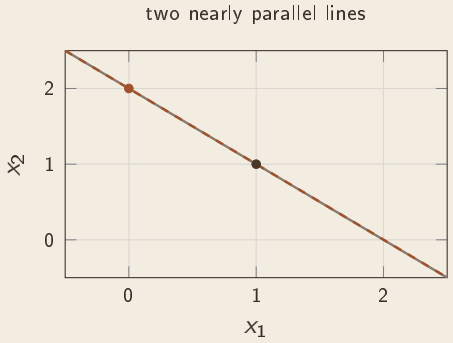
PART B

Solving linear systems

$Ax = b$ – inside every linear fit and every Newton step – and how many digits of x we can trust.

Linear systems: $A\mathrm{x} = \mathrm{b}$

Problem: find x from n linear equations in n unknowns – met as the normal equations (L3), the Newton step $J\Delta\mathrm{x} = -\mathrm{F}$ (L3) and implicit time steps (L2).



Each equation is a line; the solution is where they cross.

$$\begin{pmatrix} 1 & 1 \\ 1 & 1.001 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2.001 \end{pmatrix} \Rightarrow \mathrm{x} = (1, 1)$$

Change b_2 by 0.001, to 2.002 (dashed line):

$$\mathrm{x} = (0,\ 2).$$

A 0.05% change in b_2 moves x by 100%. The lines are nearly parallel: the problem is ill-conditioned (L1), $\kappa = 4002$.

Solving $Ax = b$ in practice

Problem: how do we solve $Ax = b$ reliably, and how many digits of x can we trust?

Elimination, not inversion

Gaussian elimination writes $A = LU$ (lower and upper triangular) and solves two triangular systems; cost $\propto n^3$ (L1). Computing A^{-1} first costs more and is less accurate: **do not invert**. `np.linalg.solve(A, b)`

How many digits survive?

Relative error in x up to $\kappa\times$ relative error in the data. With round-off $\varepsilon \approx 10^{-16}$ (L4), about $16 - \log_{10}\kappa$ correct digits remain.

Straight-line fit (L3), 10 points	$\kappa(A)$	$\kappa(A^{\mathsf{T}}A)$
$x = 0, 1, \ldots, 9$	10	100
$x = 2001, \ldots, 2010$ (years)	1.4×10^6	2×10^{12}
$x - \bar{x}$ (centred)	2.9	8

With years, about 4 digits survive. Centre x (L3), or use `lstsq` on A.

Data: none needed – κ depends only on the x values, not on the measured y.

PART C

Decomposing data

How variables vary together, and along which directions: correlation, the singular value decomposition and principal components.

Covariance and correlation

Problem: how do two variables x and y vary together, and how strongly?

Covariance:

$$s_{xy} = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})$$

positive when both tend to be above (or below) their means together; units of x times units of y.

Correlation (Pearson): $r = s_{xy}/(s_x s_y)$, between -1 and 1, no units. It measures how close the points lie to a *straight line*.

Rank correlation (Spearman), r_S: r of the ranks (L5). It measures a *monotonic* relation and is robust to bad values.

Many variables: the covariance matrix

For m variables (the columns of X): $C_{jk} = s_{x_j x_k}$; on the diagonal, the variances $s_{x_j}^2$. With centred columns, $C = X^{\mathsf{T}}X/(N-1)$.

Example, noisy van der Pol (slide 8), x_n and x_{n-4} (0.2 s apart):

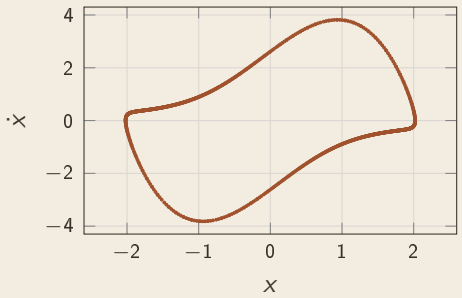
$$C = \begin{pmatrix} 2.22 & 2.16 \\ 2.16 & 2.22 \end{pmatrix}, \quad r = 0.976, \quad r_S = 0.965$$

Data: the noisy van der Pol series of slide 8.

Dependence versus association

Problem: if the correlation is zero, are the two variables unrelated?

van der Pol limit cycle: $r = 0.00$, $r_S = -0.14$



- **Association:** the two vary together. r measures straight-line association; r_S monotonic association.
- **Dependence:** knowing one changes what we know about the other. Here, given x, $\dot{x}$ can take only two values: complete dependence, zero correlation.
- A nonzero r does not show that one causes the other: both may follow a third variable, e.g. time.

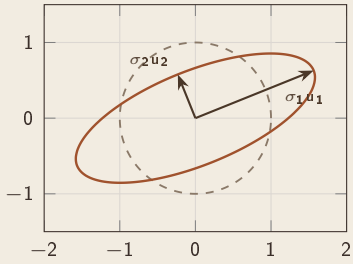
For time series, the usual p-value of r does not hold: successive values are not independent (L5).

Data: van der Pol of slide 4, 200 s, without noise.

The singular value decomposition (SVD)

Problem: how do we find the main directions of a data table – a matrix of any shape, where eigenvectors do not apply?

A maps the unit circle to an ellipse



Every $m \times n$ matrix: $\boxed{A = U\Sigma V^{\mathsf{T}}}$

- V: orthonormal (perpendicular, unit-length) input directions v_i
- Σ: **singular values** $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$, the stretches
- U: orthonormal output directions u_i

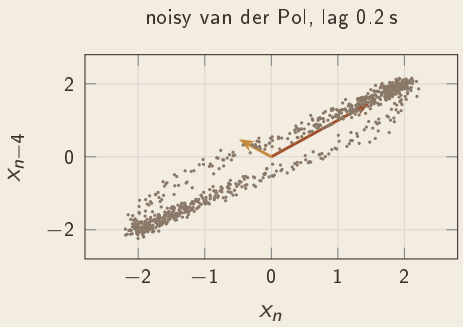
$Av_i = \sigma_i u_i$: a rotation, a stretch, a rotation.

What the σ_i tell:

- conditioning: $\kappa = \sigma_{\max}/\sigma_{\min}$ (slide 13: $2.0/0.0005 = 4002$)
- rank: the number of nonzero σ_i
- least squares: `lstsq` (L3) works through the SVD
- keep the k largest: the closest matrix of rank k (the step to PCA)

Principal component analysis (PCA)

Problem: many measured variables vary together. Along which directions do the data really vary?



Data matrix X: N rows (observations) $\times$ m columns (variables), each column minus its mean. Its SVD:

$$X = U\Sigma V^{\mathsf{T}}$$

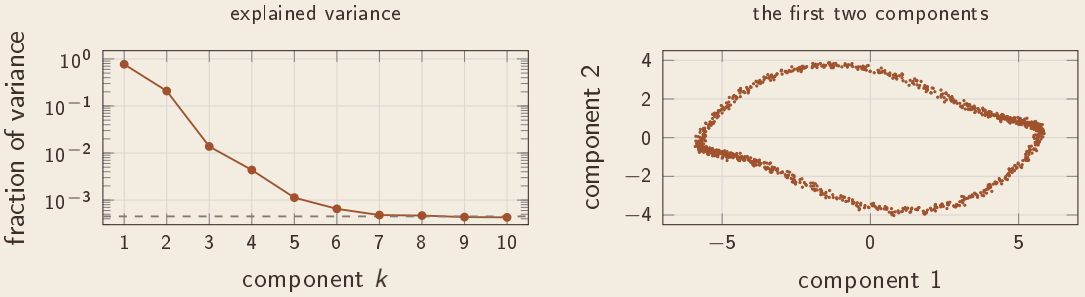
- **principal directions:** the v_k
- variance along v_k: $\sigma_k^2/(N-1)$
- **principal components:** the data in the new coordinates, XV

Left: pairs (x_n, x_{n-4}) of the noisy series of slide 8. Direction 1 (red, along the diagonal): 98.8% of the variance; direction 2 (ochre): 1.2%. Arrows: one standard deviation along each (direction 2 drawn three times longer).

Data: the noisy van der Pol series of slide 8.

How many components? Ten delay coordinates

Problem: each row is now a 1.8-s piece of the record, 10 values 0.2 s apart. How many directions hold the signal?

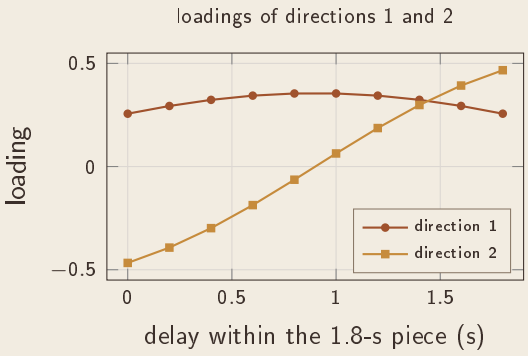


Components 1–2: 77% and 21%; 3–6: from 1.4% down to 0.07%; 7–10: a flat floor (dashed), variance $\approx 0.01 = \sigma^2$ of the added noise. Above the floor: signal; on the floor: noise. Components 1–2 trace the limit cycle; 3–6 its non-sinusoidal shape (harmonics, L6). PCA of delay coordinates is called **singular spectrum analysis**.

Data: the noisy van der Pol series of slide 8, 4001 values.

Reading the loadings

Problem: what does each principal direction mean in terms of the original variables?



The **loadings** of direction k are the entries of v_k: the weight of each original variable in that component.

Direction 1: all weights of one sign, largest in the middle – an average over the piece: where the oscillator is.

Direction 2: weights rising steadily from -0.47 to $+0.47$ – late minus early: how fast it moves.

Read the pattern of the weights, not single values; the sign of a direction is arbitrary.

Data: as on the previous slide.

Standardising: the units decide the directions

Problem: variables in different units. Does PCA depend on the units we choose?

van der Pol state, 200 s: position x and velocity $\dot{x}$

Units	Direction 1	Variance in directions 1, 2
x in m, $\dot{x}$ in m/s	mostly x	50.3%, 49.7%
x in m, $\dot{x}$ in mm/s	$\dot{x}$ alone	100%, 0%
standardised	either: no preferred direction	50.0%, 50.0%

In mm/s the velocity numbers are 1000 times larger: their variance dominates, and PCA reports one component.

Standardise

Replace each value by $z_{ij} = (x_{ij} - \bar{x}_j)/s_j$: every column has mean 0 and standard deviation 1. PCA of standardised data is PCA of the **correlation matrix**. Do it for different units or scales – not for delay coordinates (one quantity, one unit).

Data: van der Pol state $(x, \dot{x})$ of slide 4, 200 s.

What PCA does and does not tell

Problem: PCA always returns directions. What do they mean?

PCA gives	PCA does not give
straight directions of largest variance	curved structure: the limit cycle, one closed curve, needs *two* components
the number of directions above the noise floor	independent causes: components are only uncorrelated
a lower-dimensional picture of the data	importance: a weak rhythm can have small variance

Before reading any PCA: check the units (previous slide) and the noise floor (slide 20).

PART D

Outlook

Other decompositions, and the systems they suit.

Decompositions and the systems they suit

Problem: which decomposition suits which data?

Method	Idea	Assumes
PCA (today)	straight directions of largest variance	linear structure; one set of directions for the whole record
Singular spectrum analysis (today)	PCA of delay coordinates of one series	a stationary series
Dynamic mode decomposition	fit $x_{n+1} = Ax_n$ to many measured variables; each eigenvector of A is a mode with its own frequency and growth rate (slide 10)	linear, autonomous dynamics
Independent component analysis	unmix signals that are fixed mixtures of independent sources	independent, non-Gaussian sources; fixed mixing
Nonlinear mode decomposition	extract each oscillation from a time–frequency view (as the STFT, L6), keeping its harmonics in one mode; surrogates decide oscillation or noise	oscillations; frequency and amplitude may change in time
Clustering (L10)	groups of points: k-means, DBSCAN	round groups, or dense regions

Which assumption fails first?

Problem: the system types of Lectures 1 and 2 – which decompositions does each break?

System (L1, L2)	What goes wrong	Better suited
linear, autonomous, stationary (e.g. noise-driven)	little: the same straight directions throughout	PCA, SSA, DMD
linear, autonomous, decaying or growing	PCA mixes early and late	DMD (growth, decay rates)
nonlinear	a curve needs several straight directions; harmonics split into separate modes	phase space; NMD
nonautonomous	one fixed set of directions or frequencies averages the change	NMD; time-resolved views (L6)
stochastic, noisy	noise adds a floor to every component (slide 20)	components above the floor; surrogates
switching states	PCA mixes the states into one cloud	clustering (L10)
ill-conditioned (L1)	small singular values amplify errors	only well-determined components

Classify the system first (L2): the classification tells which decomposition to trust.

Which method for which question?

Question	Method	Assumes	Check
What states does it visit?	phase portrait; delay coordinates	enough coordinates; good τ	shuffled surrogate; vary τ
Solve $Ax = b$	elimination (`solve`), `lstsq`	A not singular	κ
Do two variables vary together?	r; rank r_S	straight line; monotonic	plot: $r = 0 \not\Rightarrow$ independent
Local motion at a fixed point	eigenvalues of J	close to the point	simulate
Main directions of variation	PCA (SVD of centred data)	straight directions; same units	noise floor; standardise; loadings

Before you decompose: five questions

1. What is the state of the system, and do I measure it, or must I reconstruct it? *(delay coordinates)*
2. Is the structure in the order of the values? *(compare with a shuffled surrogate)*
3. How well conditioned is my linear problem? *(κ; centre, do not invert)*
4. How many components stand above the noise floor – and is the structure straight or curved? *(PCA)*
5. Does my measure of association fit the relation? *($r = 0$ does not mean independent: plot the pair)*

Plot the points first; then choose the decomposition.

TO TAKE AWAY

In phase space a rhythm is a closed curve.

Correlation and PCA describe straight-line structure.

They can be applied when the variation lies along straight directions that stay the same through the record – a stationary system, with variables in the same units or standardised.

They cannot be applied as they stand when the structure is curved (a limit cycle), changes in time (nonautonomous, or decaying), or switches between states.